

Topics covered till Quiz I:

1. Description of a collection of atoms - Center of mass (CoM) – motion of CoM and motion relative to the CoM
2. Newton's laws of motion
3. Types of motion in one dimension possible in a given potential energy function
4. Newton's law of gravitation: 2-body problem: Effective potential energy function - bound and unbound orbits
5. Motion in central force (3D harmonic oscillator)
6. Lagrangian, Action, Principle of Stationary Action (Hamilton's principle) and Euler-Lagrangian equations
7. Phase space and Hamiltonian formalism

Coordinate systems, unit vectors in cartesian, plane polar, cylindrical polar and spherical polar coordinate systems; velocities and accelerations in these coordinate systems

Solving " $F = ma$ " for simple cases (solving 2nd order differential equations)

Conservative forces, constraints, non-conservative forces

Degrees of freedom, generalized coordinates, configurational space

Conservation of linear momentum, angular momentum, energy

Comments:

- a) There are many books dealing with Newtonian mechanics with a lot of examples (couple of them were listed in the course syllabus and resources page which I had shown in the very first few slides at the beginning of the course).
- b) So far in this course, we looked mainly at the framework of Newtonian mechanics and are comparing it with the Lagrangian framework. As part of this course, we will be going through Newtonian, Lagrangian, Hamiltonian, and then Quantum mechanical frameworks for the mechanical description of the motion of particles.
- c) One of the books that I find useful is "Introduction to Mechanics" by Kleppner and Kolenkow (this does not deal with Lagrangian mechanics)
- d) Chapter 19 of Feynman's lectures in Physics Vol.II for the Principle of Least Action (Here is the link to this chapter: https://www.feynmanlectures.caltech.edu/II_19.html)
- e) This book may be useful: <https://pressbooks.bccampus.ca/engineeringsystems/chapter/lagrangian-mechanics/>
- f) The problems chosen for the assignments are representative. Such problems with some variations will be appear in Quiz I.